

Grade 6 Accelerated
Unit 7
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

1. The formula to find the area of a trapezoid is

- ☐ $\frac{1}{2}d_1d_2$.
☐ bh .
☒ $\frac{1}{2}h(b_1 + b_2)$.

2. The formula to find the area of a rhombus is

- ☒ $\frac{1}{2}d_1d_2$.
☐ bh .
☐ $\frac{1}{2}h(b_1 + b_2)$.

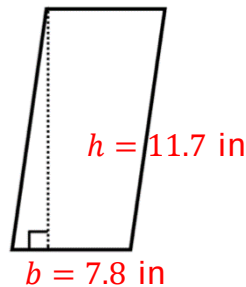
3. The formula to find the area of a parallelogram is

- ☐ $\frac{1}{2}d_1d_2$.
☒ bh .
☐ $\frac{1}{2}h(b_1 + b_2)$.

For questions 4-6, label each figure with the given dimensions, then calculate the area using the appropriate formula.

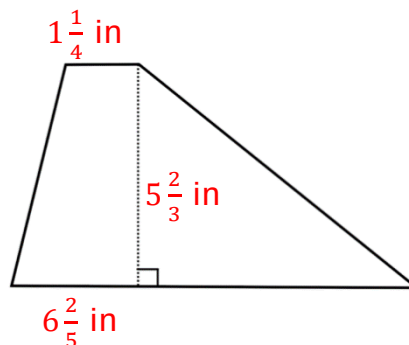
4. $b = 7.8$ in
 $h = 11.7$ in
 $7.8 \cdot 11.7$

Area = 91.26 in^2



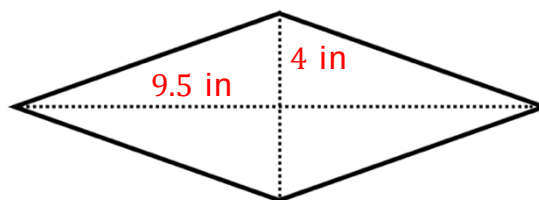
5. $b_1 = 1\frac{1}{4}$ in
 $b_2 = 6\frac{2}{5}$ in
 $h = 5\frac{2}{3}$ in
 $(1\frac{1}{4} + 6\frac{2}{5}) = 7\frac{13}{20}$ or 7.65
 $\frac{1}{2} \cdot 5\frac{2}{3} \cdot 7.65$

Area = $21\frac{37}{40}$ or 21.675 in^2



6. $d_1 = 9.5$ in
 $d_2 = 4$ in
 $\frac{1}{2} \cdot 9.5 \cdot 4$

Area = 19 in^2



7. Jocelyn made an error in her work when solving the problem below. Identify the error and then find the base of the parallelogram.

The area of a parallelogram is 75 square centimeters and the height is 6.25 centimeters. What is the length of the base of the parallelogram?

Her solution: $(75)(6.25) = 468.75 \rightarrow$ error is she multiplied the area by the height to find the base

The length of the base is 468.75 centimeters

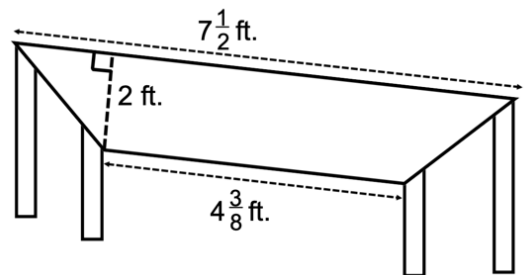
$$\text{Area} = b \cdot h$$

$$\frac{75}{6.25} = \frac{6.25b}{6.25}$$

$$b = 12 \text{ cm}$$

8. The seat of a bench in the Marlins dugout is in the shape of a trapezoid. Find the area of the bench.

$$\begin{aligned} A &= \frac{1}{2} \cdot h \cdot (b_1 + b_2) \\ &= \frac{1}{2} \cdot 2 \cdot \left(4\frac{3}{8} + 7\frac{1}{2}\right) \\ &= \frac{1}{2} \cdot 2 \cdot 11\frac{7}{8} \\ &= 11\frac{7}{8} \text{ ft}^2 \end{aligned}$$



9. The Adidas Flagship Building located in Herzogenaurach is shaped like a rhombus. One of the diagonals measures 255.4 meters and the other diagonal measures 203.7 meters. Find the area of the top of the building.

$$\begin{aligned} A &= \frac{1}{2} \cdot d_1 \cdot d_2 \\ &= \frac{1}{2} \cdot 255.4 \cdot 203.7 \\ &= 26012.49 \text{ m}^2 \end{aligned}$$

10. An eraser is in the shape of a parallelogram with a base of $1\frac{7}{12}$ inches and a height of $\frac{6}{14}$ inches. How many square inches is the top of the eraser?



$$A = b \cdot h$$

$$= 1\frac{7}{12} \cdot \frac{6}{14}$$

$$= \frac{19}{28} \text{ in}^2$$